Achieving Optimal Fraud Detection with PaySim and Random Forest

Final Examination

Zhyrgalbek Kalykov
North American University
Class: COMP 2319
Professor: Sabina Adhikari
Source Code: GitHub Repository

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Executive Summary

The rapid digitalization of financial transactions necessitates robust mechanisms for detecting anomalous and fraudulent behavior. This project details the analysis and modeling of the PaySim synthetic financial dataset, containing over 6.3 million transaction records. Due to an extreme class imbalance (less than 0.14% fraud), a systematic preprocessing pipeline was developed, utilizing One-Hot Encoding, StandardScaler normalization, and Random Undersampling. A stratified train/test split was performed *before* undersampling to ensure the test set preserved the real-world class distribution. A Random Forest Classifier was then trained on the balanced training subset and evaluated against the realistic imbalanced test set, achieving a 99.6% recall rate for fraudulent transactions and revealing the practical precision–recall tradeoff inherent in fraud detection systems.

1 Introduction and Exploratory Data Analysis (EDA)

The primary obstacle in financial fraud detection is *class imbalance*; standard algorithms tend to blindly predict the majority class (legitimate transactions) to achieve artificially high accuracy, completely missing the rare fraudulent events. Initial analysis of the 6.3 million rows revealed that fraudulent transactions were exclusively isolated to `TRANSFER` and `CASH_OUT` types and exhibited significantly higher median monetary values and volatile bursts compared to normal transactions, which were tightly clustered at lower dollar amounts.

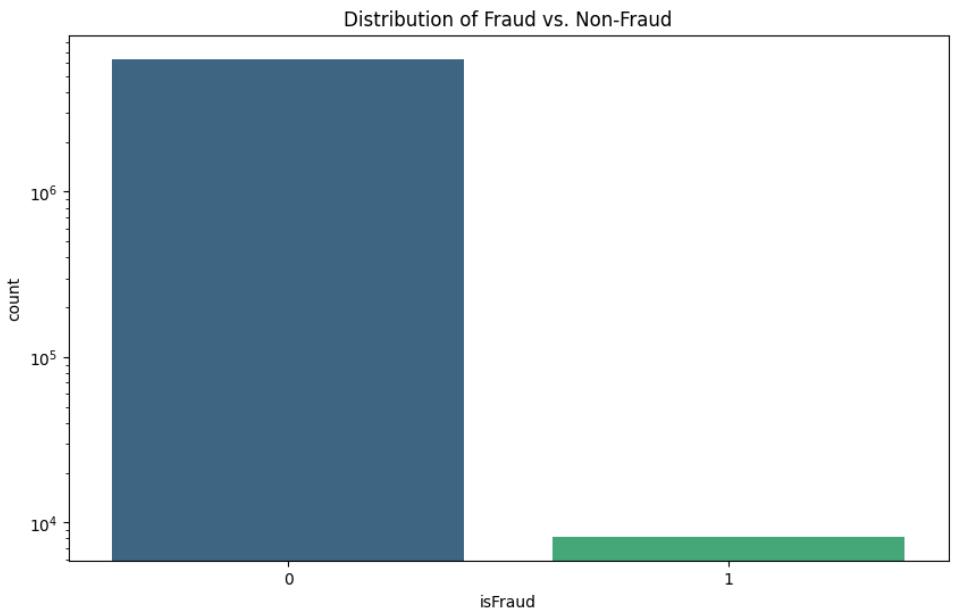

Figure 1: Original Class Distribution. Note the logarithmic scale required to visualize the rare fraud cases.

2 Methodology and Data Preprocessing

To prepare the data for the Random Forest algorithm, a multi-step transformation pipeline was executed. Critically, the dataset was split into training and testing sets

before any balancing was applied, ensuring the evaluation reflects real-world performance.

- **Categorical Encoding:** The `type` column was One-Hot Encoded (with `drop_first=True` to avoid multicollinearity) to allow independent mathematical comparison by the model.
- **Stratified Train/Test Split:** The full dataset was split 80/20 using `stratify=y`, which preserves the original 0.13% fraud rate in both the training and testing sets. This ensures the test set reflects real-world conditions.
- **Feature Scaling (Standardization):** We applied Z-score normalization (StandardScaler) to the numeric features. The scaler was fit exclusively on the training set and then used to transform both sets, preventing data leakage. Note: while Random Forests are inherently scale-invariant (they split on thresholds, not distances), scaling was applied as a best practice for pipeline consistency and potential future model comparisons.

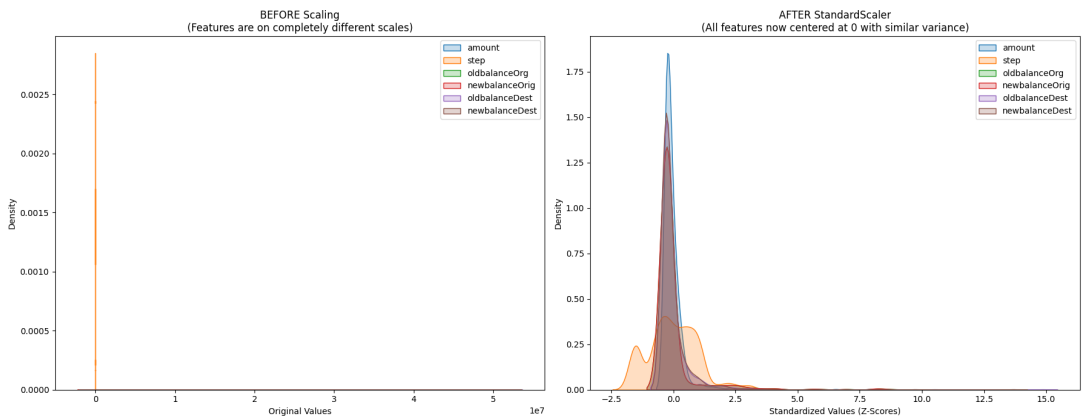

Figure 2: Feature Scaling Proof. KDE plot comparing numeric features before and after applying StandardScaler.

- **Data Balancing (Random Undersampling):** To resolve the extreme class imbalance, undersampling was applied *only to the training set*. Normal transactions were randomly downsampled to match the number of fraudulent transactions, creating a balanced training subset. The test set was left untouched at its original imbalanced distribution.

3 Model Training and Evaluation Results

The balanced training subset was used to train a Random Forest Classifier configured with 100 decision trees (`n_estimators=100`). The model was then evaluated against the full imbalanced test set of approximately 1,272,524 transactions (roughly 1,270,881 legitimate and 1,643 fraudulent).

3.1 Performance Metrics

The model achieved a **Recall of 99.6%** for the fraud class (1,637 out of 1,643 actual fraud cases detected), with only 6 fraudulent transactions missed. This is the most critical metric for a financial institution, as each missed fraud case represents direct financial loss.

However, the realistic evaluation also revealed the expected **precision–recall tradeoff**: the model flagged 16,181 legitimate transactions as suspicious (false positives), yielding a fraud precision of approximately 9.2%. In a production environment, these flagged transactions would be routed to a human review queue rather than automatically blocked—a standard and acceptable practice in the industry, where the cost of missing real fraud far outweighs the cost of investigating false alarms.

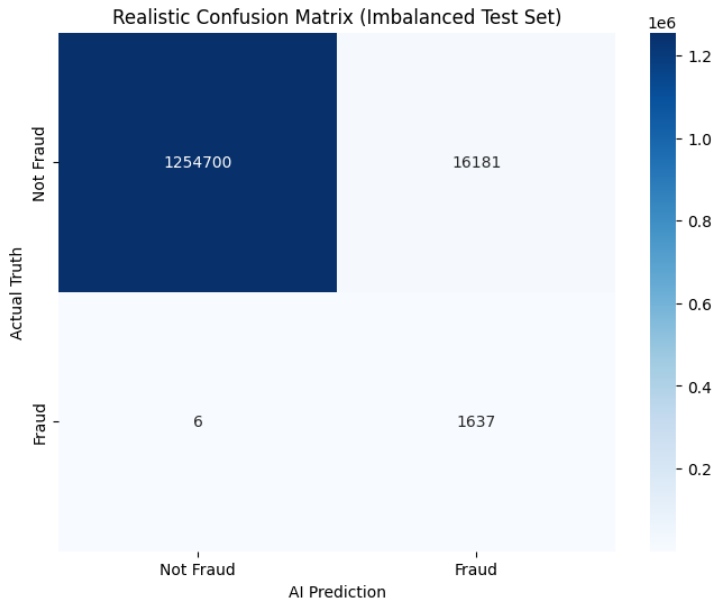

Figure 3: Realistic Confusion Matrix (Imbalanced Test Set). The model correctly identified 1,637 out of 1,643 actual fraud cases, while flagging 16,181 legitimate transactions for review.

3.2 Feature Importance

The model relied most heavily on variables related to sender and receiver account balances rather than just the raw transaction `amount`.

4 Conclusion

By establishing a robust preprocessing pipeline—specifically leveraging a stratified split before undersampling, proper scaler isolation, and training-only balancing—this project successfully transformed a massive, highly imbalanced dataset into an actionable intelligence tool. The resulting Random Forest model's 99.6% recall rate demonstrates that advanced machine learning can effectively isolate and identify complex fraud vectors, drastically mitigating financial risk. The realistic evaluation also highlights the inherent precision–recall tradeoff in fraud detection: achieving near-perfect recall requires accept-

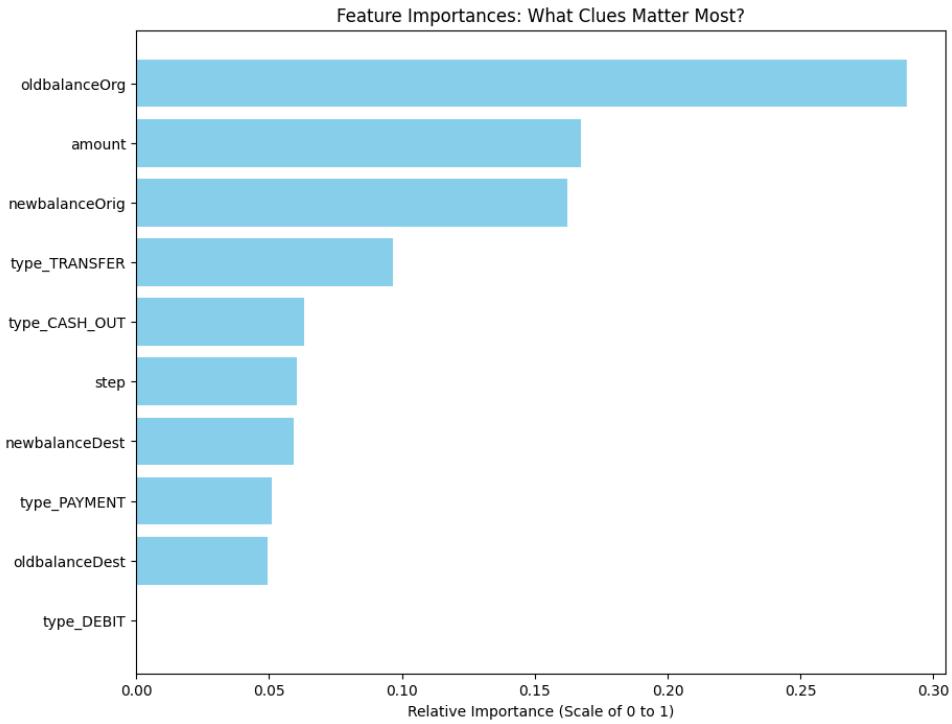

Figure 4: Feature Importance Ranking. Visual ranking of columns by their contribution to the model's decision-making process.

ing a higher false positive rate, a tradeoff that real-world financial institutions routinely manage through human review workflows.